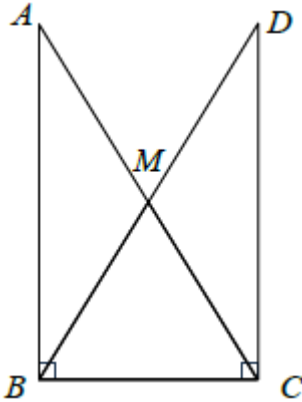
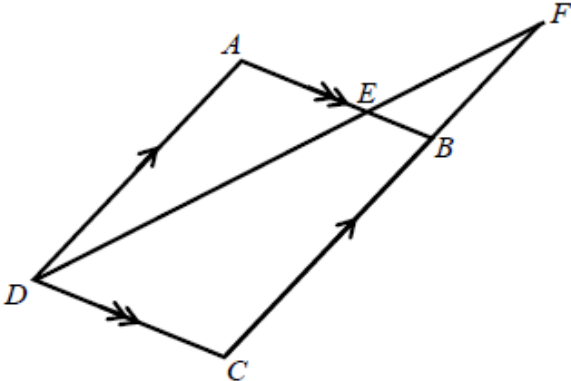
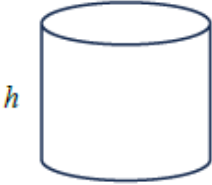
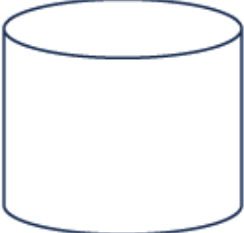
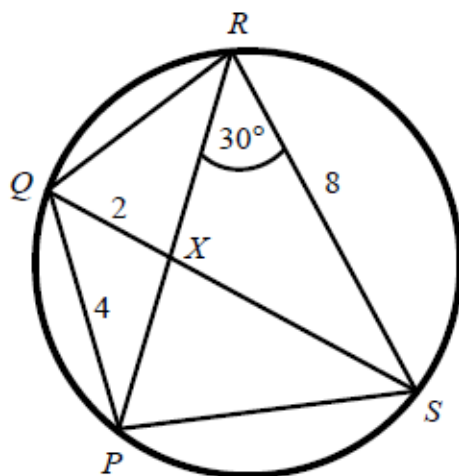


1	YTSS/2025/I/11
	In the diagram below, triangle BMC is an equilateral triangle. Angle ABC and angle DCB are 90° and M is the midpoint of AC and BD. Show that triangle ABM is congruent to triangle DCM. Give a reason for each statement you make. 
	[3]
2	SCGS/2025/I/25
	 $ABCD$ is a parallelogram. E is a point on AB such that $AE = \frac{2}{3} AB$. The line DE extended meets CB produced at F. (a) Explain why triangles ADE and BFE are similar. [2] (b) Show that $BF = \frac{1}{2} DA$. [1] (c) Find the exact value of $\frac{\text{area of triangle } BEF}{\text{area of triangle } CDF}$. [1] (d) Write down the ratio $\text{area of triangle } ADE : \text{area of quadrilateral } BCDE$. [1]

3	CCHY/2025/I/12
	Starlight Café sells latte drinks in two geometrically similar mugs. The height of the larger mug is 1.5 times the height of the smaller mug, h. Latte in a smaller mug and in a larger mug are sold at \$3.30 and \$13.20 respectively. Mandy thinks that it is more value for money to buy latte in 4 smaller mugs than one larger mug. Is Mandy correct? Justify your decision with calculations.  h smaller  $1.5 h$ larger [3]
4	GMSS/2025/I/18
	A solid cone P with a radius r cm and a height h cm has a volume of V cm³. (a) Explain the effect on the volume of cone P when its radius is doubled and the height remains unchanged. [2] (b) Cone R is geometrically similar to cone P and its height is three times that of cone P. Find, in terms of V, the volume of cone R. [2]

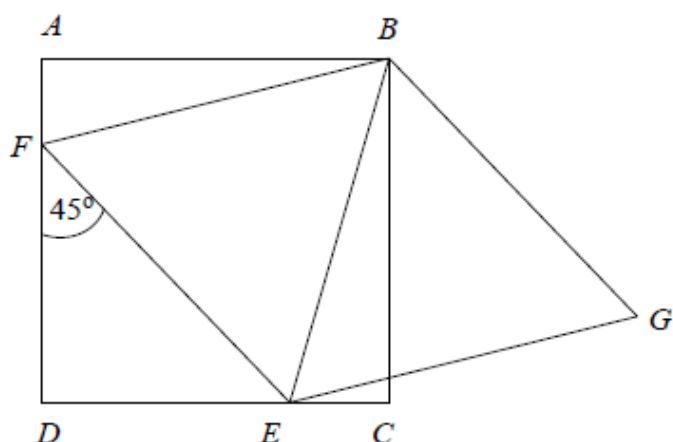
$PQRS$ is a cyclic quadrilateral and the diagonals PR and QS intersect at X .
Angle $PRS = 30^\circ$, $PQ = 4$ cm, $RS = 8$ cm and $QX = 2$ cm.



- (a) Name an angle that is equal to angle RPS and state the reason. [1]
(b) Show that triangle PQX is similar to triangle SRX .
Give a reason for each statement you make. [2]
(c) Calculate the length of RX . [2]

6 GMSS/2025/I/26

In the diagram, $ABCD$ is a square and angle DFE is 45° .



(a) Give a reason for each statement you make to show that

(i) $AF = EC$,

[2]

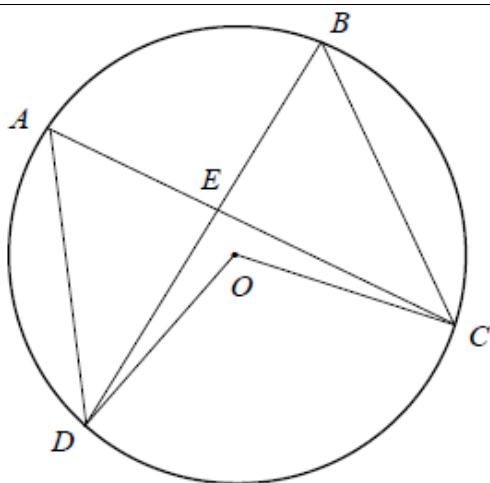
(ii) triangle BFA is congruent to triangle BEC .

[2]

(b) Given that $EGBF$ is a rhombus, calculate angle AFB .

[1]

7 ACSBR/2025/I/25



A, B, C and D are points on the circumference of the circle with centre O .

(a) Show that triangle ADE is similar to triangle BCE .

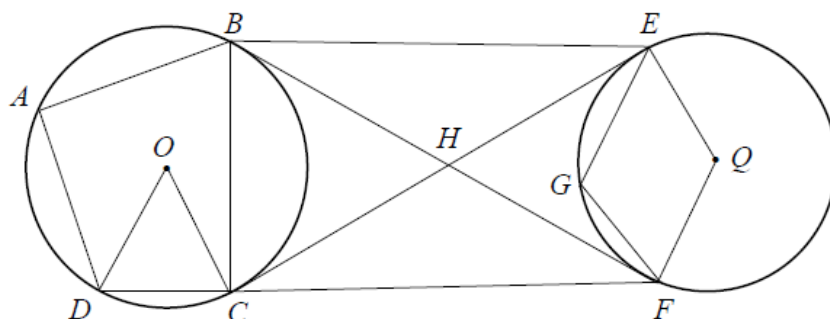
Give a reason for each statement you make.

[2]

(b) Angle $DOC = 99^\circ$ and length of minor arc $DC = 8.5$ cm.
Find the radius of the circle.

[2]

(a)



A, B, C and D are points on the circle with centre O .

E, F and G are points on the circle with centre Q .

BHF and CHE are tangents to both circles.

(i) Show that triangle BHE is congruent to triangle CHF .

Give a reason for each statement you make.

[3]

(ii) Angle $EQF = x^\circ$.

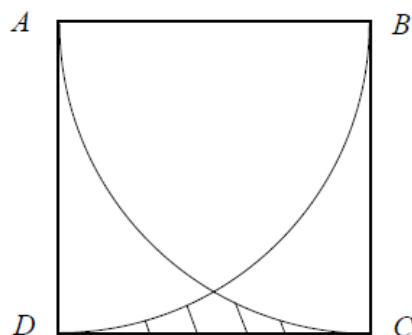
(a) Find, in terms of x , angle EGF .

[1]

(a) (ii) (b) Find, in terms of x , angle OCB .

[3]

(b)



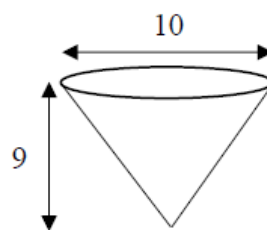
The diagram shows a square $ABCD$ with side 5 cm.

Two quadrants are drawn with each corner A and B as their centres.

Find the area of the shaded region.

[3]

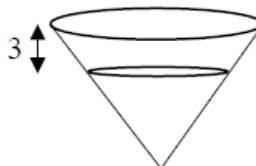
A disposable paper cup is shaped in a cone.
The diameter of the cone is 10 cm.
The height of the cone is 9 cm.
The thickness of the paper is negligible.



- (a) Calculate the curved surface area of the inside of the cup.

[3]

- (b) James filled the paper cup with water from the dispenser.
The surface of the water is 3 cm below the top of the cup.



- (i) James thinks that two-third of the inner surface area of the cup is in contact with water.

Explain why he is wrong.

[2]

- (ii) James pours the water from this paper cup into a cylindrical bottle.
The depth of water in the bottle is 0.4 cm.

Calculate the radius of the bottle.

[4]

Solution

1	$BM = CM$ (Sides of equilateral triangle) --- [B1] $AM = DM$ (M is a mid point of AC and DB) --- [B1] must explain, cannot just state $\angle AMB = \angle DCM$ (vert opp. angles) --- [B1] $\triangle ABM \equiv \triangle DCM$ (SAS) [3]
2	(a) $\hat{DAE} = \hat{FBE}$ or $\hat{ADE} = \hat{BEF}$ (alternate angles, $DA \parallel CF$) $\hat{AED} = \hat{BEF}$ (vertically opp. angles) Since there are two pairs of equal angles, triangles ADE and BFE are similar. (b) Since triangles ADE and BFE are similar, $\frac{BF}{AD} = \frac{BE}{AE} = \frac{1}{3-1} = \frac{1}{2}$ $\therefore BF = \frac{1}{2} AD$ (shown) c) $\frac{1}{9}$ d) $1 : 2$
3	$\frac{V_1}{V_2} = \left(\frac{h}{1.5h} \right)^3 = \frac{h^3}{3.375h^3}$ Volume of 4 small mugs > Volume of 1 large mug $4 > 3.375$ Cost of 4 small mugs = $\\$3.30 \times 4 = \\13.20 Cost of 1 large mug = $\\$13.20$ Mandy is correct. For the same amount paid for the latte, more latte is obtained from 4 small mugs of latte than from 1 large latte.
4	(a) $V = \frac{1}{3} \pi r^2 h$ $V_{new} = \frac{1}{3} \pi (2r)^2 h$ $V_{new} = \frac{1}{3} \pi (4r^2) h$ $V_{new} = 4V$ The volume increased by fourfold. (b) $\frac{V_R}{V} = \left(\frac{3}{1} \right)^3$ $V_R = 27V$

5	(a) Angle $\underline{SQR}$ because of <u>angles in the same segment.</u> (b) $\angle RXS = \angle PXQ$ (vert. opp. $\angle$s) $\angle SRX = \angle PQX$ (angles in the same segment) $\angle QPX = \angle RSQ$ (angles in the same segment) $\therefore \triangle PQX$ is similar to $\triangle SRX$ (c) $\frac{RX}{2} = \frac{8}{4}$ $RX = 4 \text{ cm}$						
6	(ai) $DF = DE$ (isosceles triangle) $AD = CD$ ($ABCD$ is a square) $AD - DF = CD - DE$ $\therefore AF = EC$ (aii) $AF = CE$ (from (i)), $\angle BAF = \angle ECB = 90^\circ$ $AB = BC$ ($ABCD$ is a square) $\triangle BFA$ is congruent to $\triangle BEC$ (SAS test) (b) 75°						
7	<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 10%; text-align: center; vertical-align: top;">(a)</td> <td style="padding: 5px;"> $\angle AED = \angle BEC$ (vert opp angles) $\angle DAE = \angle CBE$ (angles in the same segment) $\angle ADE = \angle BCE$ (angles in the same segment) </td> </tr> <tr> <td></td> <td style="padding: 5px;">Triangle ADE is similar to triangle BCE (AA).</td> </tr> <tr> <td style="text-align: center; vertical-align: top;">(b)</td> <td style="padding: 5px;"> $2\pi r \times \frac{99}{360} = 8.5$ $r = 4.92$ </td> </tr> </table>	(a)	$\angle AED = \angle BEC$ (vert opp angles) $\angle DAE = \angle CBE$ (angles in the same segment) $\angle ADE = \angle BCE$ (angles in the same segment)		Triangle ADE is similar to triangle BCE (AA).	(b)	$2\pi r \times \frac{99}{360} = 8.5$ $r = 4.92$
(a)	$\angle AED = \angle BEC$ (vert opp angles) $\angle DAE = \angle CBE$ (angles in the same segment) $\angle ADE = \angle BCE$ (angles in the same segment)						
	Triangle ADE is similar to triangle BCE (AA).						
(b)	$2\pi r \times \frac{99}{360} = 8.5$ $r = 4.92$						

8	(ai)	$\angle BHE = \angle CHF$ (vert opp angles)
		$HB = HC$ (tangents from ext point)
		$HE = HF$ (tangents from ext point)
		$\triangle BHE \equiv \triangle CHF$ (SAS)
	(aiia)	$\angle EGF = \frac{360 - x}{2}$
	(aiib)	$\angle EHF = 360 - 90 - 90 - x = 180 - x$ (tangent perpendicular to radius)
		$\angle EHF = \angle BHC = 180 - x$ (vert opps angles)
		$\angle BCH = \frac{180 - (180 - x)}{2} = \frac{x}{2}$ (isos triangle)
		$\angle OCB = 90 - \frac{x}{2}$
	(b)	$\frac{1}{2}(5)(5) \sin 60 = 10.825$
		$\pi(5^2) \times \frac{90 - 60}{360} \times 2 = 13.090$
		$5(5) - 10.825 - 13.090 = 1.09$

9	(a)	Slant height of cone, $\sqrt{9^2 + 5^2} = 10.296$
		Curved SA = $\pi(5)(10.296)$
		= 162cm^2
	(bi)	The ratio of area is not proportional to the ratio of heights. There is a larger surface area at the top compared to the bottom. Hence less than two-third of the inner surface area of the cup is in contact with the water.
	(bii)	$\frac{V_{Water}}{V_{Cup}} = \left(\frac{h_{Water}}{h_{Cup}}\right)^3 = \left(\frac{6}{9}\right)^3 = \frac{8}{27}$
		Volume of water in cup, $\frac{8}{27} \times \left(\frac{1}{3}\right)\pi(5^2)(9) = 69.813$
		Let the radius of the bottle be r.
		$\pi r^2(0.4) = 69.813$
		$r = 7.45$